
BDG: Batch-Dispersion Guidance for Property-Band Generation with Continuous Flow Matching

Bobo Li,¹ Henry Huang,¹ Idea Idehpour,¹ Haimo Fang¹

¹Department of Computer and Information Science, University of Pennsylvania

Abstract

A design campaign accepts every sample inside a tolerance band around a target, and we study what limits that fraction under inference-time guidance. On QM9 3D coordinates we train an E(3)-equivariant flow-matching model against a matched diffusion baseline, and confirm guidance works, roughly doubling coverage while molecule stability falls from 0.434 to between 0.199 and 0.320. The property error splits into centring and spread, and one strength dial moves both. Batch-Dispersion Guidance measures the batch spread of the guide’s endpoint prediction, forms a scale-free error against a requested setpoint, and serves the guidance term’s deviation gain by it while holding the centring gain fixed, at no additional network evaluations. The setpoint moved the guide-scored spread monotonically on three of three q50 curves, from 0.67 to 1.19 times unguided, including widening, which the strength dial does not reach. Coverage does not improve over a tuned plug-in baseline, and over 72 paired tests the largest effect is a loss: the coefficient is affine in the predicted property, so the rule is the plug-in field reweighted. Transferred unchanged to DNA enhancers on the probability simplex, the ordering reproduces and the widening does not clear noise. The setpoint orders spread at no extra sampling cost; it is not a value the sampler attains.

1 Introduction

Diffusion reverses a stochastic corruption [1] and flow matching regresses a vector field along a prescribed path [2]; stochastic interpolants contain both [3], so the backbone choice is empirical.

At a median target our batches sit 0.6 to 3.0 tolerance-band widths off target while their spread is 6.0 to 16.4 band widths, so clustering limits coverage. Guidance rules carry one strength dial that moves centring and spread together [4, 5], and we found none letting the user request a spread in property units.

We test on two state spaces whether requesting batch dispersion raises band coverage: QM9 3D coordinates, and DNA enhancers on the probability simplex. We expected the knob to raise coverage and to transfer; it transferred and the coverage gain did not appear. Our contributions are summarized as follows.

- We benchmark flow matching against published diffusion on one budget.
- We price guidance against a matched unguided control.
- BDG serves the deviation gain by the batch variance of the guide’s own endpoint prediction, at no extra network evaluations, and we state the reduction bounding it.
- The controller transfers to the simplex, where we measure the same signatures.

2 Related Work

Continuous models and guidance. Flow matching [2], rectified flow [6] and EDM [1] are endpoints of one construction [3, 7]. Our comparators are DPS [8], LGD-MC [9], TMPD [10], TFG [11] and D-Flow [12].

Near our mechanism. Prior work drives ensemble moments to a target [13], tilts ensemble variance toward wider batches [14], sets ensemble spread by the step count [15], couples trajectories repulsively [16], widens through negative guidance [17], and reads the guidance scale as a control gain [18, 19]. In a search we do not claim is exhaustive we did not find a weight set by the batch variance of a *learned predictor’s* endpoint prediction against a *two-sided user-chosen* setpoint.

The two modalities. EDM defines the stability conventions we adopt [20], and property-conditioned 3D generation is our task [21–24]. On the simplex, Dirichlet [25], Fisher [26] flow matching and MOG-DFM [27] score the task by Fréchet biological distance at 500 base pairs; Gumbel-Softmax flow matching [28] works the simplex on other tasks. We score GC-band coverage at 200 base pairs (Appendix A.4).

3 Methods

3.1 Modalities, Data, and Preprocessing

Modality 1 is QM9, 133,885 molecules on $\mathbb{R}^{N \times 3}$: coordinates zero-centred, a hard one-hot over {H,C,N,O,F}, no feature scaling. Splits are `train_a` for the generator and guide f_A , `train_b` for the evaluator f_B , and `val` and `test` held out (Appendix A.1). We generate unconditionally; the properties are dipole μ (D), polarizability α (Bohr³) and HOMO-LUMO gap (Ha).

Modality 2 is 6,126 *Drosophila* Kenyon-cell enhancer regions [27] on $(\Delta^3)^L$ at $L=200$, decoded by per-position arg max: no rotation group, and an exact property.

3.2 Continuous Flow Matching Baseline

With $t : 0 \rightarrow 1$ from noise to data, the linear interpolant $x_t = tx_1 + (1-t)x_0$ has constant conditional velocity $u_t = x_1 - x_0$, and the source is projected onto the zero-centre-of-mass subspace of dimension $3(N-1)$. The objective is a masked mean over $t \sim \mathcal{U}(0, 1)$, coordinates and types normalised separately, $\lambda = 1$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\frac{1}{3\Sigma M} \|M \odot (v_\theta^c - u^c)\|^2 + \lambda \frac{1}{K\Sigma M} \|M \odot (v_\theta^f - u^f)\|^2 \right], \quad (1)$$

with M the atom mask, ΣM its sum, K the type channels and the superscripts c, f the coordinate and type heads. The network is a dense E(3)-equivariant graph network of 3.75 M parameters; Table 6 gives its architecture and optimiser settings. We sample with explicit Euler in 100 steps.

3.3 Diffusion Baseline

The diffusion baseline shares everything with Section 3.2 but the corruption, which makes Section 4.2 interpretable: a variance-preserving process with $\beta(\tau) = \beta_{\min} + \tau(\beta_{\max} - \beta_{\min})$, $\beta_{\min} = 0.1$, $\beta_{\max} = 20$:

$$x_\tau = \alpha(\tau)x_0 + \sigma(\tau)\epsilon, \quad \log \alpha(\tau) = -\frac{1}{2}(\beta_{\min}\tau + \frac{1}{2}(\beta_{\max} - \beta_{\min})\tau^2), \quad \alpha^2 + \sigma^2 = 1. \quad (2)$$

The network predicts ϵ . We integrate $dx/d\tau = -\frac{1}{2}\beta(\tau)(x + s_\theta)$ backwards, $s_\theta = -\epsilon_\theta/\sigma$, on a decreasing 100-step grid, then jump to the Tweedie mean.

3.4 Guided Generation

Each guided step toward target y forms the endpoint estimate m_i , being $x_t^{(i)} + (1-t)v_\theta(x_t^{(i)}, t)$ for the flow model and the Tweedie mean for the diffusion model, evaluates $F_i = f_A(m_i)$ and $g_i = \nabla_m f_A(m_i)$, and adds

$$G_i = \frac{\text{num}_i}{s^2} J_i^\top g_i, \quad \text{num}_i^{\text{plug}} = y - F_i, \quad (3)$$

Table 1: Base models on QM9, unconditional, 100-step Euler ODE, our evaluator under EDM conventions. Our flow row is three seeds of 10,000 samples, $\pm$ one seed sd; EDMsecond is borrowed, $n=2,000$, one seed, so it is not comparable. — is pending (Appendix A.7).

Model	Source	Atom stab. $\uparrow$	Mol. stab. $\uparrow$	Validity $\uparrow$	Uniq. $\uparrow$	NFE $\downarrow$	Params
Flow matching (ours)	reproduced	0.9366 $\pm$.0012	0.3993 $\pm$.0085	0.7625 $\pm$.0031	0.9939	100	3.75 M
VP diffusion (ours)	reproduced	—	—	—	—	101	3.75 M
EDMsecond [20]	borrowed	0.9644	0.6725	0.8490	—	100	5.34 M
EDM [20]	quoted	0.9870	0.8200	0.9190	—	1000	—
Real QM9 (ceiling)	measured	0.994	0.956	0.982	—	—	—

Table 2: Guided against unguided at q50, otherwise matched. Cells are μ / gap / α ; bold is the best guided arm in a column, a dash a cell we did not run. Every guided arm falls below the 0.390 chemistry floor. TFG is ported, not tuned here (Appendix A.4).

Arm	IN-BAND $\uparrow$	Spread/unguided	Mol. stab. $\uparrow$
unguided	0.090/0.121/0.051	1.00/1.00/1.00	0.434/0.434/0.434
plug-in (best w)	0.184/0.219/ 0.133	0.52/0.60/0.47	0.246/0.199/0.320
TFG ($w=4$, μ only)	0.535 /—/—	0.22/—/—	0.172/—/—
BDG ($w=32$)	0.180/ 0.227 /0.106	0.49/0.68/0.53	0.348 /0.262/0.340

Table 3: BDG ablations on μ at q50: gates, ladder, controls. τ_{mult} scales the setpoint by the unguided batch’s spread, so 1.0 requests it; rungs 0.75 and 1.25 and the other two properties are in Table 8. $w=1$, floor 0.390.

Variant	Role	η	τ_{mult}	Spread/ung.	IN-BAND $\uparrow$	Mol. stab. $\uparrow$
plug-in, $w=1$	base	—	—	0.714	0.1094	0.3633
BDG $\eta=0$	gate	0	1.0	0.714	0.1094	0.3633
BDG one-sided, expand	gate	4	1.5	0.714	0.1094	0.3633
BDG contract	ladder	4	0.5	0.670	0.1406	0.3359
BDG neutral	ladder	4	1.0	0.900	0.0820	0.4258
BDG expand	ladder	4	1.5	1.093	0.0703	0.4141
plug-in, $w=4$	control	—	—	0.584	0.1406	0.2656
open-loop e replay	control	4	0.5	0.686	—	—
signed- w plug-in	control	—	—	—	—	—

report IN-BAND on decoded molecules and the spread ratio against unguided, at the **q50** target, the median of the training property distribution. The *chemistry floor* is $0.9\times$ the unguided arm’s molecule stability, and we do not rank an arm below it. Arms share the checkpoint, guide, evaluator, target, atom counts and noise, and Table 2 reports each arm’s best strength. Spread ratios are scored by f_A and IN-BAND by the held-out f_B . Table 1 uses three seeds; every other cell uses one at $n=256$, where the standard error on IN-BAND is about 0.02.

4.2 Modality 1: Flow Matching versus Diffusion

In Table 1 our flow model reached 0.9366 atom and 0.3993 molecule stability at 100 evaluations, against EDM’s published 0.9870 and 0.8200. We did not test the causes of that gap; EDM’s own ablation of unscaled one-hot features loses 3 atom and 35 molecule points [20]. We carry flow matching forward without a fidelity win, because the diffusion row is pending, and take the endpoint estimate from its linear path.

4.3 Modality 1: Guidance Works

Guidance raised IN-BAND by $2.0\times$ on μ , $1.8\times$ on the gap and $2.6\times$ on α for the plug-in rule, and TFG reached $5.9\times$ on μ (Table 2). TFG took the most coverage here and the least chemistry, which is the trade-off BDG is built to make requestable.

4.4 Modality 1: Innovation Ablations

The gates passed exactly in Table 3: at $\eta = 0$ and in the one-sided variant at an expanding setpoint, BDG reproduced the plug-in field to zero. The setpoint moved spread monotonically on three of three q50 curves, 0.670 to 1.187 times unguided (Table 8, Figure 2), and the deviation gain changed sign within a trajectory. Coverage did not improve. Every rung clearing the chemistry floor is nominally worse than plug-in, and the largest effect over 72 paired tests is a BDG *loss* at $z = -3.07$; the grid is correlated, so we read it as a null. Plug-in at $w = 4$ reaches BDG’s contraction rung at the same

Table 4: Four recent guidance rules ported to our frozen checkpoint, guide and evaluator at q50, $w=1$. No result cell has run. Passes are network calls at batch granularity per cell of $n=5,000$; the generator column differs across arms, so read the two together.

Method	Status	IN-BAND $\uparrow$	Mol. stab. $\uparrow$	Diversity $\uparrow$	Gen. passes	Guide passes
DPS-style plug-in [8]	reproduced	—	—	—	10,000	4,000
TMPD-inspired [10]	reproduced	—	—	—	10,000	4,000
LGD-MC [9]	reproduced	—	—	—	12,000	20,000
TFG [11]	reproduced	—	—	—	8,000	20,000
Flow matching, unguided	internal	—	—	—	10,000	0
BDG (ours)	ours	—	—	—	10,000	4,000

Table 5: Transfer to DNA enhancers on the simplex, $L=200$. We score GC-band coverage, not these papers’ FBD at 500 base pairs, so their numbers are never pooled with ours.

Signature	QM9	DNA simplex
$\eta=0$ reproduces plug-in	bit-identical	bit-identical
one-sided at an expanding setpoint	exact	exact, $w_{\text{eff}} = +1.000$
setpoint ladder monotone in spread	$0.67 \rightarrow 1.19$	$0.81 \rightarrow 1.01$
contraction breaks the chemistry floor	yes	yes
IN-BAND improves	no	no
<i>Against recent simplex sequence models</i>		
Dirichlet FM [25]	—	—
Fisher Flow [26]	—	—
MOG-DFM [27]	—	—
BDG, transferred (ours)	—	—

IN-BAND, on lower molecule stability, a difference of $z = 1.73$ that one seed does not resolve and that sits below the floor either way; and the open-loop replay recovers 69.5% to 96% of the closed loop’s spread shift, so the schedule and not the feedback carries most of it.

4.5 Modality 1: Comparison with Recent Methods

We ported four guidance rules to one frozen generator (Table 4); Appendix A.4 gives three port caveats. BDG makes one guide forward and one backward per guided step, the plug-in rule’s exact call pattern.

4.6 Modality 2: Transfer of the Innovation

BDG transferred with no change to the controller and coverage did not improve (Table 5). The ladder’s range is 1.7 to $2.6\times$ smaller, and its widening rung reaches only 1.006 to 1.008 , inside the noise at this n , so the ordering transfers and the widening does not. GC is affine in the simplex coordinates and averages over 200 positions.

4.7 Cross-Modality Analysis and Failure Modes

The direction matched across modalities and the magnitude did not, which the property map accounts for. The trust region acts at large $|w_{\text{eff}}|$: in Table 2’s cells the plug-in rule clips 701 sample steps where BDG clips 4,457, so a rung can be clip-limited and not controller-limited. And this grid cannot separate chemistry cost from $|w_{\text{eff}}|$ and τ_{mult} . Figure 5 shows what contraction breaks.

5 Discussion

Guidance worked at a measured chemistry cost, and the ablations support one mechanism claim beside one null on coverage. Both transferred to the simplex.

Equation (5) accounts for the null. BDG reaches nothing outside the plug-in field’s two-parameter family, and we swept one of those parameters. The one chemistry gain we measure, $+0.07$ at $n=256$, is unresolved at $z = 1.73$.

Four limitations bound the claim: the trust region confounds the largest rungs, the signed- w control has not run, the open-loop control rests on one seed pair, and both cells of the chemistry comparison sit below our floor. The narrowest claim supported is monotone, two-sided, modality-portable spread control at no extra sampling cost, the setpoint ordering spread without attaining it.

References

- [1] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” *Advances in neural information processing systems*, vol. 35, pp. 26565–26577, 2022.
- [2] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” *arXiv preprint arXiv:2210.02747*, 2022.
- [3] M. Albergo, N. M. Boffi, and E. Vanden-Eijnden, “Stochastic interpolants: A unifying framework for flows and diffusions,” *Journal of Machine Learning Research*, vol. 26, no. 209, pp. 1–80, 2025.
- [4] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [5] Q. Zheng, M. Le, N. Shaul, Y. Lipman, A. Grover, and R. T. Chen, “Guided flows for generative modeling and decision making,” *arXiv preprint arXiv:2311.13443*, 2023.
- [6] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” *arXiv preprint arXiv:2209.03003*, 2022.
- [7] N. Ma, M. Goldstein, M. S. Albergo, N. M. Boffi, E. Vanden-Eijnden, and S. Xie, “Sit: Exploring flow and diffusion-based generative models with scalable interpolant transformers,” in *European Conference on Computer Vision*, pp. 23–40, Springer, 2024.
- [8] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. C. Ye, “Diffusion posterior sampling for general noisy inverse problems,” in *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.14687.
- [9] J. Song, Q. Zhang, H. Yin, M. Mardani, M.-Y. Liu, J. Kautz, Y. Chen, and A. Vahdat, “Loss-guided diffusion models for plug-and-play controllable generation,” *Proceedings of the 40th International Conference on Machine Learning (PMLR v202)*, 2023. <https://proceedings.mlr.press/v202/song23k.html>.
- [10] B. Boys, M. Girolami, J. Pidstrigach, S. Reich, A. Mosca, and O. D. Akyildiz, “Tweedie moment projected diffusions for inverse problems,” *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2310.06721.
- [11] H. Ye, H. Lin, J. Han, M. Xu, S. Liu, Y. Liang, J. Ma, J. Zou, and S. Ermon, “Tf: Unified training-free guidance for diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2409.15761.
- [12] H. Ben-Hamu, O. Puny, I. Gat, B. Karrer, U. Singer, and Y. Lipman, “D-flow: Differentiating through flows for controlled generation,” in *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.14017.
- [13] E. Lempereur, N. Cuvelles-Magar, F. Coeurdoux, S. Mallat, and E. Vanden-Eijnden, “MGD: Moment guided diffusion for maximum entropy generation,” *arXiv preprint arXiv:2602.17211*, 2026.
- [14] I. Azangulov, L. Zhang, and K. Ashouritaklimi, “Variance-tilted diffusion models for diverse sampling,” *arXiv preprint arXiv:2606.22239*, 2026.
- [15] F. Merizzi, D. Evangelista, and H. Loukos, “Controlling ensemble variance in diffusion models: An application for reanalyses downscaling,” *arXiv preprint arXiv:2501.14822*, 2025.
- [16] G. Corso, Y. Xu, V. De Bortoli, R. Barzilay, and T. Jaakkola, “Particle guidance: Non-I.I.D. diverse sampling with diffusion models,” in *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.13102.
- [17] E. Ventura, B. Achilli, L. Ambrogioni, and C. Lucibello, “Negative guidance reduces means and expands variances in diffusion models,” *arXiv preprint arXiv:2602.00716*, 2026.

- [18] H. Wang, Y. Liu, J. Chi, F. Liu, R. Xue, and Y. Duan, "CFG-Ctrl: Control-based classifier-free diffusion guidance," *arXiv preprint arXiv:2603.03281*, 2026.
- [19] F. Koulischer, F. Handke, J. Deleu, T. Demeester, and L. Ambrogioni, "Feedback guidance of diffusion models," *arXiv preprint arXiv:2506.06085*, 2025.
- [20] E. Hoogeboom, V. G. Satorras, C. Vignac, and M. Welling, "Equivariant diffusion for molecule generation in 3d," in *International conference on machine learning*, pp. 8867–8887, PMLR, 2022.
- [21] F. Bao, M. Zhao, Z. Hao, P. Li, C. Li, and J. Zhu, "Equivariant energy-guided SDE for inverse molecular design," *International Conference on Learning Representations (ICLR)*, 2023.
- [22] M. Xu, L. Yu, Y. Song, C. Shi, S. Ermon, and J. Tang, "Geodiff: A geometric diffusion model for molecular conformation generation," *arXiv preprint arXiv:2203.02923*, 2022.
- [23] R. Irwin, A. Tibo, J. P. Janet, and S. Olsson, "Semlaflow—efficient 3d molecular generation with latent attention and equivariant flow matching," *arXiv preprint arXiv:2406.07266*, 2024.
- [24] Y. Song, J. Gong, M. Xu, Z. Cao, Y. Lan, S. Ermon, H. Zhou, and W.-Y. Ma, "Equivariant flow matching with hybrid probability transport for 3d molecule generation," *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [25] H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, and T. Jaakkola, "Dirichlet flow matching with applications to dna sequence design," *arXiv preprint arXiv:2402.05841*, 2024.
- [26] O. Davis, S. Kessler, M. Petrache, I. I. Ceylan, M. Bronstein, and A. J. Bose, "Fisher flow matching for generative modeling over discrete data," *Advances in Neural Information Processing Systems*, vol. 37, pp. 139054–139084, 2024.
- [27] T. Chen, Y. Zhang, S. Tang, and P. Chatterjee, "Multi-objective-guided discrete flow matching for controllable biological sequence design," *arXiv preprint arXiv:2505.07086*, 2025.
- [28] S. Tang, Y. Zhang, A. Tong, and P. Chatterjee, "Gumbel-softmax flow matching with straight-through guidance for controllable biological sequence generation," *ArXiv*, pp. arXiv–2503, 2025.

A Appendix

A.1 Dataset and Preprocessing Details

QM9 is the DeepChem packaging, pinned by SHA-256 on the archive, both source files and the built tensor. We keep the 3,054 uncharacterized molecules that EDM and its successors drop, a disclosed deviation. Splits come from one seeded permutation with $n_{\text{test}} = 13,083$ and $n_{\text{val}} = 17,748$, the remainder split in half. Normalisation constants from `train_a`: μ 2.70376 $\pm$ 1.53941 D, α 75.21779 $\pm$ 8.20041 Bohr³, gap 0.25126 $\pm$ 0.04746 Ha. The size histograms of the two training halves have total variation 0.0064 with mean atom counts 18.003 and 17.985, which makes them interchangeable as atom-count sources. Equivariance is verified numerically: 12 of 12 checks pass to $\sim 10^{-17}$ for rotation, translation, permutation and padding, on both the scalar and velocity heads. Modality 2 uses all 6,126 Kenyon-cell sequences with no train/validation split, because the evaluator is an exact count and no leakage is possible.

A.2 Training and Sampling Details

Guide and evaluator are E(3)-invariant networks of width 128 and 4 layers, 429,825 parameters, trained 120 epochs with Adam at 3×10^{-4} , cosine decay, batch 128, MSE on z -scored targets, selected on best validation MAE. Measured validation MAE on 17,748 molecules: f_A 0.0897 D, 0.2447, 0.00390 Ha; f_B 0.0840, 0.2407, 0.00380. Both are less accurate than published predictors because they are undertrained relative to them, and since δ scales with the evaluator’s MAE our bands are correspondingly wider, which is a disclosed conservatism. The sampler sweep at fixed checkpoint gives atom stability 0.9334, 0.9388, 0.9399, 0.9394 at NFE 100, 250, 500 and 1000, so

Table 6: The three trained networks. Only the generator is large. Both predictors are frozen at sampling time and neither is updated by guidance. Values are read from the checkpoints’ own recorded arguments.

Setting	Generator v_θ	Guide f_A	Evaluator f_B
Role	samples	steers	scores
Used at sampling time	yes, 1 fwd/step	yes, 1 fwd + 1 bwd/guided step	no
Architecture	E(3)-equivariant EGNN	E(3)-invariant EGNN	E(3)-invariant EGNN
Width $\times$ depth	256×8	128×4	128×4
Parameters	3.75 M	429,825	429,825
Training split	<code>train_a</code> (51,527)	<code>train_a</code> (51,527)	<code>train_b</code> (51,527)
Objective	linear-interpolant FM	MSE on z -scored y	MSE on z -scored y
Optimiser	Adam	Adam	Adam
Learning rate	2×10^{-4} , cosine to 5%	3×10^{-4} , cosine	3×10^{-4} , cosine
Epochs / batch	1500 / 256	120 / 128	120 / 128
EMA decay	0.9999	none	none
Selection	final EMA weights	best validation MAE	best validation MAE
Seed	20260918	—	—
Wall-clock	11.4 h (303k steps)	—	—
Checkpoint md5	a190ac83...	—	—

Table 7: Sampler settings per arm. Every arm shares one frozen generator, one frozen guide, one 100-step Euler grid and one trust region, so arms differ only in the guidance rule and its constants. A dash marks a constant the arm does not have. — marks a setting the v3 run fixes.

Arm	w	η	τ_{mult}	t_{min}	Batch B	Seeds	n per cell
unguided	—	—	—	—	256	1	256
plug-in	1	—	—	0.5	256	1	256
plug-in, strength sweep	4	—	—	0.5	256	1	256
BDG, $\eta=0$ gate	1	0	1.0	0.5	256	1	256
BDG, setpoint ladder	1	4	0.5–1.5	0.5	256	1	256
BDG, one-sided gate	1	4	1.5	0.5	256	1	256
BDG, high strength	32	4	0.5	0.5	256	1	256
v3 main run, all arms	—	—	—	0.5	—	3	5,000
Modality 2, all arms	—	—	—	—	—	—	—

quality saturates by 500 steps. Table 6 gives every trained component and Table 7 every sampler setting per arm.

Hardware and wall-clock for the sampling arms are recorded per cell in the run manifests and are summarised in Table 11.

A.3 Additional Ablations

Table 8 is the full five-rung setpoint ladder behind the three rungs quoted in the main text, on all three properties, with the controller’s own run-mean diagnostics beside the sample statistics. It is the measurement that the monotonicity claim rests on. Table 9 reserves the v3 extension of the same sweep.

Two facts in this table are worth separating. The spread column is monotone in τ_{mult} on all three properties, which is the control result. The IN-BAND column is not ordered and its best cell is the contraction rung, which also carries the lowest molecule stability on μ and gap, which is the fidelity cost. Neither the mean V_b/τ^2 nor w_{eff} settles at the equilibrium value $\tau^2(1 - 1/\eta)$ within the 50 guided steps, so these are transient, not converged, operating points.

A.4 Baseline Fidelity and Modality 2 Disclosures

Modality 1 naming. Our plug-in arm is a DPS-style Gaussian-energy field, where the original DPS differentiates the residual L_2 norm. Our TMPD arm replaces the constant denominator by $s^2 + g^\top \Sigma g$, which is the published method’s scalar affine-property limit, included as a related-work comparison rather than a faithful nonlinear port. TFG runs at its published QM9 hyperparameters through our deterministic ODE sampler where the original used a stochastic DDIM update, and our unscaled

Table 8: The complete setpoint ladder at $\eta=4$, $w=1$, q50 target, one seed, $n=256$ per cell. Spread is $\text{sd}(f_A)$ over the batch divided by the unguided arm’s. V_b/τ^2 and w_{eff} are run means over guided steps. The two gate rows are exact: $\eta=0$ is bit-identical to the plug-in arm on all three properties, and the one-sided variant at an expanding setpoint holds $w_{\text{eff}}=+1.000$. **Resolution:** at $n=256$ the standard error on IN-BAND is about 0.019 near $p=0.1$, so IN-BAND differences below roughly two points are not resolvable here, and the BDG rows’ error is understated because the batch is coupled through $\bar{F}$ and V_b . The ordering of the spread column, not the size of any single IN-BAND difference, is what this table establishes.

Property	Arm	τ_{mult}	Spread / ung.	V_b/τ^2	w_{eff}	IN-BAND $\uparrow$	Mol. stab. $\uparrow$
μ	unguided	—	1.000	—	—	0.0898	0.4336
μ	plug-in	—	0.714	—	+1.000	0.1094	0.3633
μ	BDG	0.50	0.670	1.2731	+2.093	0.1406	0.3359
μ	BDG	0.75	0.813	0.9351	+0.741	0.0820	0.4102
μ	BDG	1.00	0.900	0.7578	+0.031	0.0820	0.4258
μ	BDG	1.25	0.982	0.6746	−0.302	0.0742	0.4219
μ	BDG	1.50	1.093	0.6459	−0.416	0.0703	0.4141
α	unguided	—	1.000	—	—	0.0508	0.4336
α	plug-in	—	0.849	—	+1.000	0.0547	0.3711
α	BDG	0.50	0.674	1.5365	+3.146	0.0742	0.3711
α	BDG	0.75	0.844	1.0761	+1.305	0.0547	0.3633
α	BDG	1.00	1.014	0.8777	+0.511	0.0625	0.3711
α	BDG	1.25	1.099	0.7190	−0.124	0.0547	0.3672
α	BDG	1.50	1.187	0.6166	−0.533	0.0430	0.3945
gap	unguided	—	1.000	—	—	0.1211	0.4336
gap	plug-in	—	0.871	—	+1.000	0.1562	0.3359
gap	BDG	0.50	0.742	1.3621	+2.448	0.1406	0.3086
gap	BDG	0.75	0.893	1.0024	+1.009	0.1055	0.3672
gap	BDG	1.00	0.968	0.7781	+0.113	0.0977	0.3906
gap	BDG	1.25	1.052	0.6376	−0.450	0.1016	0.3594
gap	BDG	1.50	1.073	0.5401	−0.840	0.0938	0.3203

Table 9: Reserved: the v3 extension of the ablation, three seeds and two backends at $n=5,000$. Every cell reports IN-BAND with its seed standard deviation and molecule stability. Row and column structure, the grid and the ordering are final; only the values are pending. This grid exists to separate the two confounds the one-seed table above cannot: whether monotonicity survives seed noise, and whether the gain η and the nominal strength w act independently.

η (at $w=1$)	τ_{mult}				
	0.50	0.75	1.00	1.25	1.50
0	—	—	—	—	—
1	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—
8	—	—	—	—	—
η (at $w=4$)	0.50	0.75	1.00	1.25	1.50
0	—	—	—	—	—
1	—	—	—	—	—
2	—	—	—	—	—
4	—	—	—	—	—
8	—	—	—	—	—

one-hot features move roughly 16 times less relative to coordinates than the released checkpoint’s, so its numbers here must never be read against its published table.

Modality 2 deviations. The standard enhancer protocol is class-conditional generation at 500 base pairs on 81-class or 47-class datasets of roughly 90,000 sequences, scored by Fréchet biological distance over 10,000 samples per model. We use GC-band coverage instead of class-conditional distance, 6,126 sequences instead of 83,968, and 200 base pairs in the shipped cells. Three further facts bear on the comparison. Gumbel-Softmax flow matching does not run the enhancer benchmark at all, reporting promoter, protein and peptide tasks, so only two of the three closest named methods apply here. Fisher Flow disputes the standard evaluator, reporting that the cell-type classifiers behind it score 11.5% and 11.2% test accuracy. MOG-DFM’s own guided demonstration runs at length 100 while reporting base distances at 500, so short-length guided evaluation is precedented. At 200 base pairs the reference classifier global-average-pools and returns meaningless numbers rather than failing loudly, which is a further reason we do not quote it.

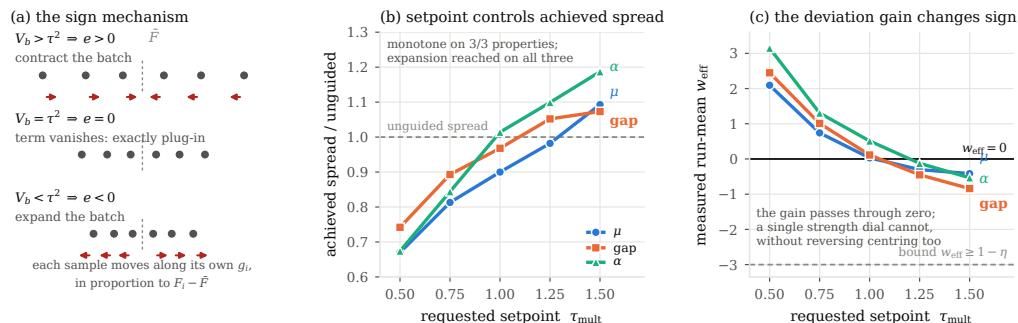


Figure 2: The controller, from the measurements in Table 8. Left: the rule. The deviation gain is servoed while the centring gain is held at w , so the step is the plug-in field at a rescaled weight aiming at a shifted target. Middle: the setpoint ladder. Requested spread against achieved spread on all three properties, monotone on all of them, with the unit line marking the unguided spread. Right: the signed effective gain w_{eff} along the ladder. It crosses zero, which is what makes the control two-sided: the same rule contracts and expands depending on the sign of the dispersion error.

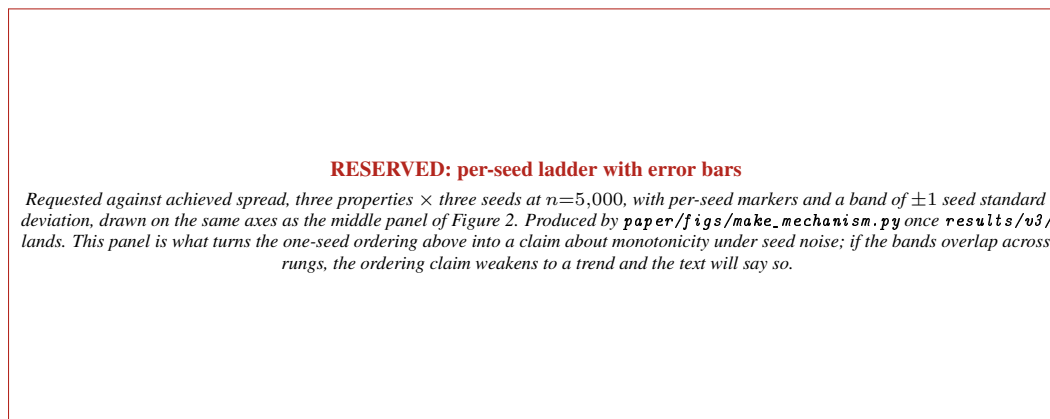


Figure 3: Reserved: monotonicity under seed noise.

Per-seed values. Table 10 reserves the per-seed breakdown behind every aggregate in the main text. Three seeds is the minimum at which a standard deviation means anything, and with $n=5,000$ per cell the standard error on IN-BAND near $p=0.1$ is about 0.004, roughly five times finer than the one-seed ablation above. Seed-to-seed spread is reported for every arm, including the arms whose aggregates we do not claim a difference for, so that a reader can check whether a difference we do claim exceeds it.

Table 10: Reserved: per-seed IN-BAND and molecule stability for every arm in Tables 2 to 4 of the main text, $n=5,000$ per cell, q50 target. Rows, columns and the seeds are final; only the values are pending. The last column is the quantity every main-text comparison must clear.

Arm	Property	Seed 1	Seed 2	Seed 3	Mean	Seed sd
unguided	$\mu / \alpha / \text{gap}$	—	—	—	—	—
plug-in	$\mu / \alpha / \text{gap}$	—	—	—	—	—
TMPD-inspired	$\mu / \alpha / \text{gap}$	—	—	—	—	—
LGD-MC	$\mu / \alpha / \text{gap}$	—	—	—	—	—
TFG	$\mu / \alpha / \text{gap}$	—	—	—	—	—
D-Flow	$\mu / \alpha / \text{gap}$	—	—	—	—	—
BDG (ours)	$\mu / \alpha / \text{gap}$	—	—	—	—	—

A.5 Additional Samples and Failure Cases

Figures 4 and 5 are the qualitative counterpart to Table 8. Their purpose is adversarial: the contraction rung is the cell with the best IN-BAND and the worst molecule stability, so it is the cell whose samples most need to be shown rather than summarised.

RESERVED: molecule grid across the setpoint ladder

A 5×8 grid of decoded molecules, one row per ladder rung $\tau_{\text{mult}} \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ at $\eta=4$, $w=1$, on μ , drawn from a fixed seed so rows are paired sample for sample. Each molecule is annotated with its f_B property value, whether it falls in the band, and whether it passes the EDM stability check. Unstable molecules are kept in the grid and marked, not filtered out.

Figure 4: Reserved: samples along the setpoint ladder.

RESERVED: failure cases at the contraction rung

The eight lowest-stability molecules from the $\tau_{\text{mult}}=0.5$ cell beside their paired unguided samples from the same seed and noise draw, with the per-atom valency violations marked. Below them, eight generated DNA sequences at the matching contraction rung with their GC content and the band edges. This panel exists to show what the fidelity cost in Table 8 actually looks like.

Figure 5: Reserved: what the contraction rung breaks.

A.6 Algorithms

Algorithm 1 One guided sampling step with Batch-Dispersion Guidance. Nothing is trained at sampling time; the generator v_θ and guide f_A are frozen. Modality-specific inputs are the endpoint map and the property f_A .

Require: batch $x_t \in \mathbb{R}^{B \times d}$, time t , strength w , gain η , setpoint τ , scale s , step h

- 1: $v_i \leftarrow v_\theta(x_t^{(i)}, t)$ ▷ one generator pass, shared with the unguided sampler
- 2: **if** $t < t_{\min}$ **then return** $x_t + h v$ ▷ guidance window
- 3: **end if**
- 4: $m_i \leftarrow x_t^{(i)} + (1 - t) v_i$ ▷ endpoint estimate
- 5: $F_i \leftarrow f_A(m_i)$; $g_i \leftarrow \nabla_m f_A(m_i)$ ▷ one guide forward, one backward
- 6: $\bar{F} \leftarrow \frac{1}{B} \sum_i F_i$; $V_b \leftarrow \frac{1}{B-1} \sum_i (F_i - \bar{F})^2$ ▷ the batch is the estimator
- 7: $e \leftarrow (V_b - \tau^2)/\tau^2$; **if** one-sided **then** $e \leftarrow \max(e, 0)$
- 8: $\text{num}_i \leftarrow (y - F_i) - \eta e (F_i - \bar{F})$ ▷ $= (y - \bar{F}) - w_{\text{eff}}(F_i - \bar{F})$, $w_{\text{eff}} = 1 + \eta e$
- 9: $G_i \leftarrow (\text{num}_i/s^2) J_i^\top g_i$ ▷ one vector-Jacobian product
- 10: $G_i \leftarrow w \frac{1-t}{t} G_i$; $G_i \leftarrow G_i \cdot \min(1, \|v_i\|/\|G_i\|)$ ▷ trust region, shared by all arms
- 11: **return** $x_t + h(v + G)$

If $B < 2$ the unbiased variance is undefined, and the implementation sets $V_b := \tau^2$ so that $e = 0$ and the step degrades to the plug-in rule rather than emitting a non-finite value.

A.7 Implementation and Reproducibility Details

Every number in this paper is produced by one script reading one recorded artefact, listed in Table 11. The innovation itself is the `mode == "bdg"` branch of `proj1/src/guidance.py`, gated by 25 checks in `proj1/tests/test_bdg.py`, which include the two exactness properties asserted in Table 8.

Table 11: Provenance. Each row names the object in the paper, the script that emits it and the recorded artefact that script reads. `—` marks an artefact the v3 run has not written. Nothing in the paper is assembled by hand.

Object	Produced by	Reads
Table 1, base models	<code>scripts/benchmark_base.py</code>	<code>weights/fm_ema.pt, results/bench/</code>
Table 2, guidance arms	<code>scripts/onegen_full_metrics.py</code>	<code>results/bdg_local/</code>
Table 3, BDG ablation	<code>scripts/bdg_full_metrics.py</code>	<code>results/bdg_port/cells/</code>
Table 4, recent methods	<code>scripts/transfer_sweep.py</code>	<code>results/v3/ —</code>
Table 5, Modality 2	<code>m2/m2_sweep.py</code>	<code>results/m2/ —</code>
Table 8	<code>scripts/bdg_table.py</code>	<code>results/bdg_port/table.txt</code>
Table 9	<code>scripts/bdg_table.py</code>	<code>results/v3/ —</code>
Figure 1, overview	<code>paper/figs/overview.tex</code>	nothing; it is a diagram
Figure 2	<code>paper/figs/make_mechanism.py</code>	<code>results/bdg_port/table.txt</code>
Figures 3–5	<code>paper/figs/make_mechanism.py</code>	<code>results/v3/ —</code>
The innovation	<code>proj1/src/guidance.py, mode == "bdg"</code>	—
Its gates	<code>proj1/tests/test_bdg.py (25 checks)</code>	—

Versions and identifiers. The generator checkpoint is pinned by md5 `a190ac83...` and QM9 by SHA-256 on both the source archive and the built tensor. Software versions, the remaining checkpoint hashes and the per-cell run manifests are `—` until the v3 environment is frozen; they are recorded by the same scripts that emit the tables above, so they land with the numbers rather than being transcribed after the fact.